Security-Aware Federated Reinforcement Learning for Resilient Recovery of Cyber-Physical Systems Under Cross-Layer Attacks

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***Abstract*— Cross-layer attacks against cyber-physical systems (CPS) can corrupt sensing, communication, and control channels at the same time, forcing recovery controllers to restore safe operation with partial observations and untrusted participants. Existing reinforcement learning (RL) recovery methods are still largely centralized, which exposes operational trajectories and scales poorly when subsystems are geographically distributed. This paper proposes a security-aware heterogeneous federated reinforcement learning (FRL) framework for collaborative resilient recovery under cross-layer attacks. The framework decomposes a CPS into subsystem-level agents, trains local recovery policies with prioritized multi-agent RL, and exchanges model updates without sharing raw trajectories. Its core mechanism, Advantage Execution–Federated Critic (AEFC), combines local safety adapters, recovery-state prototype alignment, high-advantage sparse update transmission, and shift-aware credibility aggregation. Local adapters keep subsystem-specific sensing and actuation private, prototype alignment stabilizes the shared recovery representation, sparse transmission limits communication to recovery-informative updates, and credibility aggregation reduces the impact of stale, noisy, or adversarial contributions. AEFC is positioned as a communication-efficient and empirically privacy-enhancing recovery mechanism, not as a formal differential-privacy or secure-aggregation protocol. The framework is evaluated on IEEE 39-bus and IEEE 118-bus power-grid CPS benchmarks implemented through Simulink–Python co-simulation. In these benchmarks, AEFC-FRL improves recovery accuracy by up to 15.3 percentage points over centralized MADDPG and reduces communication overhead by 45.8% compared with standard FedRL, while maintaining stronger resilience under malicious updates. The evaluation also reports heterogeneity stress tests, membership-inference risk, gradient-leakage limitations, and deployment constraints for practical CPS recovery.**

***Index Terms*—Cyber-physical systems, cross-layer attacks, resilient recovery, security-aware federated reinforcement learning, multi-agent reinforcement learning.**

I. INTRODUCTION

CYBER-PHYSICAL systems (CPS) tightly couple computation, communication, sensing, and physical processes, and they now support critical infrastructure such as power grids, transportation systems, industrial automation, and healthcare platforms. Their operational value comes with rising security risk. A cross-layer attacker may corrupt sensor streams, delay or drop network messages, and manipulate control commands within the same incident. Such attacks do not only degrade prediction accuracy; they can push the physical plant toward unsafe operating regions before a human operator can isolate the compromised component. For this reason, post-attack resilient recovery is a first-order security problem in CPS, not merely a performance optimization problem.

This paper focuses on collaborative recovery after cross-layer attacks. The goal is to restore safe operation under partial observability, heterogeneous subsystem dynamics, and untrusted or unreliable participants. Timely recovery is therefore as important as prevention, especially when the system must reconfigure control actions while the attack surface remains active. Fig. 1 summarizes the multi-layer recovery setting considered in this work.

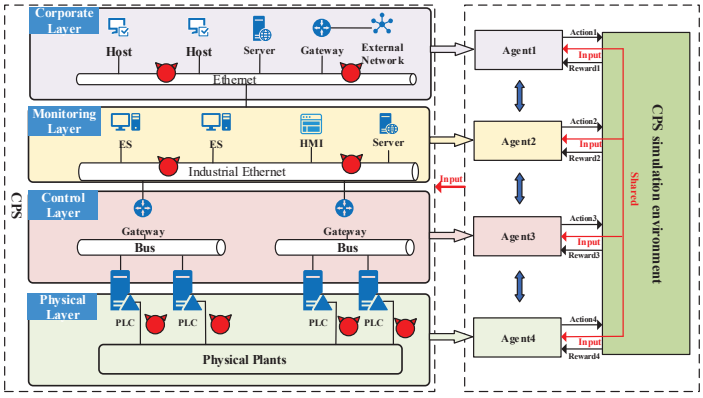


Fig. 1: The multi-layer resilient recovery framework.

Reinforcement learning (RL) is well suited to resilient recovery because it optimizes sequential control decisions through interaction with complex environments. Existing RL-based recovery methods, however, are still largely centralized. They often require collecting trajectories or intermediate statistics from several subsystems, which can expose operational states and attack traces. They also scale poorly when local dynamics, sensors, and actuators differ across geographically distributed components.

Federated reinforcement learning (FRL) offers a natural alternative: each subsystem learns from local trajectories and exchanges model information with a coordinating server. This formulation preserves data locality, but data locality is not a formal privacy guarantee. Unless differential privacy, secure aggregation, or cryptographic protection is added, shared gradients and sparse updates may still leak rare local states or attack trajectories. We therefore treat privacy as an empirical risk dimension, while positioning resilient recovery under cross-layer attacks as the main security objective.

Conventional FRL is not enough for this setting. Under cross-layer attacks, the most useful recovery transitions are often rare, and their parameter changes need not be the largest coordinates. Magnitude-only Top-κ compression can keep large coordinates caused by load variation or corrupted observations while dropping lower-magnitude coordinates associated with actions that return the plant to a safe region. Dense FedAvg has a different weakness:

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it averages stale, compromised, or observation-incompatible updates as if they came from one stationary task. Credibility weighting also remains weak if the score reflects only client identity or update norm rather than recovery value and representation drift.

To address these failure modes, we develop a security-aware collaborative FRL framework for CPS resilient recovery. The CPS is decomposed into subsystem-level agents trained with prioritized multi-agent reinforcement learning. On top of this local learner, the proposed Advantage Execution–Federated Critic (AEFC) mechanism changes how existing FL primitives are used. Sparse upload is selected by recovery advantage rather than by magnitude alone. Shared parameters are restricted to an aligned recovery backbone after local adapters absorb subsystem-specific observations and actions. Server-side weights are computed from critic-informed recovery utility and prototype drift, so large but stale or inconsistent updates are down-weighted. AEFC is therefore a recovery-conditioned sparse sharing and aggregation rule, not a generic compressed FedRL pipeline.

The learning setting is heterogeneous FRL rather than vertical federated learning in the usual sense of feature-partitioned data with aligned sample identities. Different CPS subsystems collect different local observations and trajectories, while their policies share only the recovery backbone after local observation and action adaptation. This distinction determines how parameter sharing, convergence analysis, and privacy claims should be interpreted.

The present work makes four contributions.

- We formulate cross-layer CPS resilient recovery as a subsystem-level cooperative multi-agent reinforcement learning problem and train local controllers with a prioritized replay variant of MADDPG, improving sample efficiency on critical recovery transitions.
- We design a heterogeneous FRL collaboration mechanism that keeps raw trajectories local while synchronizing only an aligned recovery backbone. Local adapters handle subsystem-specific observation and action spaces, and recovery-state prototypes provide a compact alignment signal for heterogeneous clients.
- We propose the AEFC optimization mechanism, which replaces magnitude-only update compression with recovery-advantage-guided sparse transmission and combines it with prototype-based recovery alignment and shift-aware credibility aggregation. This design targets the failure modes of directly applying generic Top-κ compression and parameter averaging to attack-time CPS recovery.
- We provide a conditional convergence analysis under smoothness, bounded heterogeneity, contractive compression, and bounded credibility-bias assumptions, and we evaluate the framework on IEEE 39-bus and IEEE 118-bus power-grid CPS benchmarks with performance, communication, heterogeneity stress, Byzantine-robustness, empirical privacy-risk, deployment-oriented analyses, and a small non-grid traffic-signal proof of concept.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the system model, adversary model, local multi-agent learner, federated training mechanism, and the AEFC optimization. Section IV reports the experimental results and discusses external validity and limitations. Section V concludes the paper.

II. RELATED WORK

Research on CPS recovery has developed along three closely related lines: resilient control design, reinforcement learning for sequential CPS decisions, and federated learning for distributed CPS data and models.

A. Resilient Design for CPS

Resilient design in CPS is commonly discussed from two perspectives: recovery strategies and recovery mechanisms [1], [2]. Recovery strategies define the target operating state and the restoration path after a disruption, whereas recovery mechanisms provide the control, optimization, or learning techniques used to realize that path.

Prior work distinguishes shallow recovery from deep recovery [3]. Shallow recovery usually addresses localized disturbances, such as abnormal sensor readings, transient communication delays, or single-component failures, and is often implemented through redundancy, component substitution, or local repair. Deep recovery targets wider disruptions, including controller compromise, large network outages, or severe data loss, and typically requires longer-horizon optimization. Duan et al. [4], for example, proposed a two-layer network recovery strategy that decomposes a large network into core and peripheral subnetworks and restores them through a multi-level mechanism.

Many early recovery mechanisms rely on control theory or game theory. Robust control, observer reconstruction, and verification-based designs have improved resilience under structured uncertainty and attacks [5]–[8]. Liang et al. [9] developed a learning-based passive resilient controller that confines the system state within a secure ellipsoidal region under covert deception attacks. Zhong et al. [10] proposed a cascaded recovery model that coordinates absorption, survival, and recovery. Cheng et al. [11] modeled infrastructure defense as a two-person, three-stage game, and Yu et al. [12] studied a modeling-control-design framework based on multi-agent game theory.

These methods provide useful recovery principles, but they are often tailored to specific models, single-scenario assumptions, or small-scale settings. Large CPS introduce data silos, heterogeneous local dynamics, and cross-layer coupling, which make purely centralized or model-specific designs harder to deploy. This motivates learning-based recovery methods that can adapt from interaction data while respecting distributed operation constraints.

B. Reinforcement Learning for CPS

Reinforcement learning has been used in CPS because it can optimize sequential decisions without requiring a complete analytical model of the environment [13], [14]. Yan et al. [15] studied replay attack detection with model-free reinforcement learning and treated attackers as proactive agents. Su et al. [16] combined self-organizing maps with deep reinforcement learning for multivariate anomaly detection. Xue et al. [17] formulated anomaly-threshold selection as a Markov decision process and solved it with deep reinforcement learning. Liu et al. [18] introduced an R-MDP formulation for deadline-aware CPS tasks and designed rewards that encourage timely behavior.

Because CPS are distributed and tightly coupled, a single learning agent may be unable to capture subsystem interactions. Multi-agent reinforcement learning (MARL) is therefore a natural fit for coordinated CPS control [19], [20]. Abiad et al. [21] developed a MARL-based caching strategy for collaborative throughput optimization. Digital-twin-based multi-agent architectures have also been used for monitoring and anomaly detection [22]. Wang et al. [23] modeled cross-layer routing in dynamic mobile networks as a decentralized partially observable Markov decision process and

used graph neural network-based reinforcement learning. In transportation CPS, Xi et al. [24] proposed a MARL cluster-control algorithm for collaborative underwater control with centralized training and distributed execution.

Despite these advances, RL and MARL still face practical barriers in CPS recovery. Centralized training can require large volumes of operational trajectories, which is undesirable when subsystems are owned by different entities or connected through bandwidth-limited links. Frequent dense communication among agents can also strain the control network. These issues motivate a federated formulation in which learning signals are shared without exposing raw trajectories.

C. Federated Learning for CPS

Federated learning allows clients to train a shared model while keeping raw data local. Its known challenges include statistical heterogeneity, system heterogeneity, client drift, communication cost, and privacy leakage from shared updates [25], [26]. These issues are amplified in CPS because measurements are operationally sensitive, distributed, and generated by safety-critical physical processes. Recent CPS-oriented studies have explored federated sensing, transportation, energy scheduling, maritime systems, and industrial edge intelligence [27]–[30].

Two bodies of FL work are most relevant to this paper. The first concerns secure and communication-efficient aggregation. ESFL improves backdoor resistance through efficient secure filtering [31], while Byzantine-robust and privacy-preserving schemes use blockchain-assisted coordination, malicious-client detection, TEEs, oblivious learning, or differential privacy [32]–[35]. Other CPS and edge-FL studies reduce communication through adaptive aggregation, edge-cloud resource allocation, incentive mechanisms, over-the-air aggregation, or compression [36]–[40]. These methods provide important infrastructure, but they usually optimize supervised or generic training objectives. In attack-time CPS recovery, the value of an update also depends on whether it encodes an action that restores safety.

The second body concerns heterogeneous and privacy-aware FL. Personalized and representation-aligned methods, including FedARA, Fed-ADE, FedDAP, and F^{2}DC, separate shared and local components or use prototypes to handle client-specific domains [41]–[44]. Privacy-focused methods such as DP-FedAdamW, FedMOP, and personalized federated diffusion training further show that data locality and sparsification are not substitutes for differential privacy, secure aggregation, or comparable mechanisms [45]–[47]. These studies inform the design and the privacy boundary of AEFC-FRL, but they do not solve sequential resilient recovery under coupled physical, sensing, communication, and control-layer attacks.

Federated reinforcement learning (FRL) extends FL to sequential decision-making and has been studied for distributed control, intelligent networking, malware mitigation, machine unlearning, predictive maintenance, and energy-system scheduling [29], [48]–[51]. CyberForce is particularly relevant because it evaluates poisoning resilience in an IoT security setting [49]. Runtime detection for backdoored DRL agents also shows that learned policies can become attack surfaces [52]. These works motivate secure FRL, but most do not address cross-layer CPS recovery with heterogeneous observation/action spaces and recovery-time communication limits.

We therefore position this work as heterogeneous FRL rather than vertical federated learning. Classical vertical FL assumes feature-partitioned data with aligned sample identities [53]; here, agents interact with different local environments and generate different trajectories. AEFC-FRL shares only an aligned recovery backbone, while local adapters keep subsystem-specific observation and action mappings private. Compared with existing CPS-oriented FL and FRL methods, the key difference is not any single primitive. Top-κ sparsification, representation alignment, and credibility weighting all have precedents. AEFC-FRL changes their optimization target: the sparse mask depends on recovery advantage, the shared representation is aligned through recovery-state prototypes, and the aggregation weight combines critic-informed utility with drift. This makes the method a security-aware recovery mechanism rather than a generic compression scheme applied unchanged to CPS.

III. METHODOLOGY

The proposed heterogeneous federated reinforcement learning (FRL) framework consists of a recovery model, a local MARL learner, a federated training procedure, and the AEFC update mechanism. The convergence analysis is stated under explicit assumptions to avoid treating non-stationary MARL training as standard convex optimization.

A. System Model

We model the CPS as a continuous-time nonlinear system. Let $x(t) \in \mathbb{R}^n$ denote the physical state at time t, and let $u(t) \in \mathbb{R}^m$ denote the control input. In nominal operation, the dynamics are given by $\dot{x}(t) = f(x(t), u(t))$, where $f(\cdot)$ describes the underlying physical process. To represent cross-layer disruptions, we introduce an attack vector $a(t)$ with components $a_p(t)$, $a_n(t)$, and $a_c(t)$ acting on the physical, network/sensing, and control layers, respectively. The attacked dynamics and observation model are

$$\begin{cases} \dot{x}(t) = f\big(x(t), u(t) + \Delta u(t)\big) + \Delta x(t), \\ y(t) = h(x(t)) + \Delta y(t), \end{cases} \tag{1}$$

where $h(\cdot)$ is the observation function. The perturbations are defined as $\Delta x(t) = D_p(a_p(t))$, $\Delta u(t) = D_c(a_c(t))$, and $\Delta y(t) = D_s(a_n(t))$. The functions $D_p(\cdot)$, $D_c(\cdot)$, and $D_s(\cdot)$ specify how attack signals are injected into the physical dynamics, control channel, and sensing/communication channel. This representation covers both single-layer disturbances and coordinated cross-layer attacks.

The recovery goal is to return the system to a safe operating region after an attack while limiting recovery time, state deviation, and control effort. Let $\mathcal{X}_{\text{safe}}$ denote the safe region and let $\phi(x_t, \mathcal{X}_{\text{safe}})$ measure the distance or violation degree of the current state relative to that region. We use an RL formulation and define a reward that penalizes unsafe deviation and excessive control. A representative form is

$$r(s_t, a_t) = -\phi(x_t, \mathcal{X}_{\text{safe}}) - \lambda \|a_t\|_2^2, \tag{2}$$

where $\lambda > 0$ controls the action penalty. The objective is to maximize the discounted return

$$J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right], \tag{3}$$

where $\gamma \in [0, 1)$ is the discount factor, s_t is the global CPS state, $a_t = (a_{1,t}, \ldots, a_{N,t})$ is the joint action, and π is the joint recovery policy. Under the above reward, maximizing $J(\pi)$ corresponds to reducing accumulated state deviation and action cost during recovery. Eq. 1 and Eq. 3 define the continuous-state, continuous-action recovery problem considered in this paper.

B. Adversary Model

We consider a recovery-oriented threat model in which the CPS may face coordinated cross-layer attacks and the federated learning process may include unreliable or malicious clients. At the physical and control layers, the adversary can perturb state evolution, sensor measurements, communication timing, or actuation commands through the attack vector in Eq. 1. At the federated layer, the coordinating server is assumed to be honest-but-curious: it follows the prescribed aggregation protocol but may inspect uploaded sparse updates, public model snapshots, and aggregate statistics. The server does not receive raw local trajectories or local observation/action adapters.

Among the participating clients, up to 30% may be Byzantine, consistent with the robustness experiments in Section IV. Byzantine clients may submit random, scaled, stale, or sign-flipped updates to slow down convergence or bias the recovery policy. They cannot access other clients' raw trajectories, private adapters, replay buffers, or the server-side aggregation logic. We also consider inference adversaries that use uploaded index-value pairs, sparse gradients, and public model snapshots to perform membership inference or gradient reconstruction. We do not consider attacks that fully compromise the global server, break cryptographic primitives, or bypass certified local protection logic. Under this model, AEFC-FRL targets resilient recovery and empirical privacy-risk reduction; it does not claim formal privacy guarantees without additional mechanisms such as secure aggregation or differential privacy.

C. MARL Training Algorithm

We decompose the CPS into N subsystem agents. Agent i observes a local state $o_i(t)$, selects a local recovery action $a_i(t) \in \mathcal{A}_i$, and receives the global resilience reward $r(t)$. The joint state is denoted by $s(t)$, and the joint action is $a(t) = (a_1(t), \ldots, a_N(t))$. We adopt centralized training with decentralized execution: during training, each critic can use the joint state and joint action to stabilize learning; during execution, each actor uses only its own local observation.

The local learner is based on MADDPG [54]. Each agent maintains an actor $\pi_{\theta_i} : o_i \mapsto a_i$ and a critic $Q_{\omega_i}(s, a_1, \ldots, a_N)$. The replay buffer of agent i is denoted by $\mathcal{D}_i$ and stores transitions (s, a, r, s'). To improve sample efficiency on rare but important recovery events, we use prioritized experience replay (PER). For transition j, the priority is

$$p_j = (|\delta_i^j| + \epsilon)^\alpha, \tag{4}$$

where δ_i^j is the TD error, $\epsilon > 0$ avoids zero priority, and α controls the strength of prioritization. The sampling probability is $P(j) = p_j / \sum_k p_k$, and the importance-sampling weight is $w_j = (|\mathcal{D}_i| P(j))^{-\beta}$, where β is annealed toward 1.

At each update, agent i samples a minibatch $\mathcal{B}_i = \{(s^j, a^j, r^j, s'^j)\}_{j=1}^B$ from $\mathcal{D}_i$. The critic loss is

$$L_{Q_i}(\omega_i) = \frac{1}{B} \sum_{j=1}^{B} w_j \left[Q_{\omega_i}(s^j, a_1^j, \ldots, a_N^j) - y_i^j \right]^2, \tag{5}$$

with target

$$y_i^j = r^j + \gamma Q_{\omega_i^-}(s'^j, a_1'^j, \ldots, a_N'^j), \tag{6}$$

where $a_k'^j = \pi_{\theta_k^-}(o_k'^j)$ is computed by the target actor of agent k, and $Q_{\omega_i^-}$ is the target critic. The actor is updated by minimizing

$$L_{\pi_i}(\theta_i) = -\frac{1}{B} \sum_{j=1}^{B} w_j Q_{\omega_i}\left(s^j, a_1^j, \ldots, \pi_{\theta_i}(o_i^j), \ldots, a_N^j\right). \tag{7}$$

Algorithm 1: MADDPG with Prioritized Replay (PER-MADDPG)

Input: Number of agents N, discount factor γ, learning rates $\alpha_\theta, \alpha_\omega$, replay buffer size, batch size B, prioritization parameters (α, β)
Output: Trained actor policies $\{\pi_{\theta_i}\}_{i=1}^N$

Initialize for each agent i: actor π_{θ_i}, critic Q_{ω_i}, target networks $\pi_{\theta_i^-}$, $Q_{\omega_i^-}$ with $\theta_i^- \leftarrow \theta_i$, $\omega_i^- \leftarrow \omega_i$, and replay buffer $\mathcal{D}_i$;
for *episode* $= 1$ **to** M **do**
  Reset the environment to s_0; each agent observes o_i;
  for $t = 0$ **to** T **do**
    for $i = 1$ **to** N **do**
      $a_i \leftarrow \pi_{\theta_i}(o_i) + \mathcal{N}_i$;
    Execute $a = (a_1, \ldots, a_N)$ and observe r, s', and local observations $\{o_i'\}_{i=1}^N$;
    Store (s, a, r, s') in each $\mathcal{D}_i$;
    for $i = 1$ **to** N **do**
      Sample $\mathcal{B}_i$ from $\mathcal{D}_i$ with probability $P(j) \propto (|\delta_i^j| + \epsilon)^\alpha$;
      Compute y_i^j by Eq. 6 and update Q_{ω_i} by Eq. 5;
      Update π_{θ_i} using the policy gradient implied by Eq. 7;
      Update priorities $p_j \leftarrow (|\delta_i^j| + \epsilon)^\alpha$;
      Soft-update target networks: $\theta_i^- \leftarrow \tau\theta_i + (1-\tau)\theta_i^-$, $\omega_i^- \leftarrow \tau\omega_i + (1-\tau)\omega_i^-$;
    $s \leftarrow s'$;

Fig. 2 illustrates the local PER-MADDPG learner used by each subsystem before federated coordination.

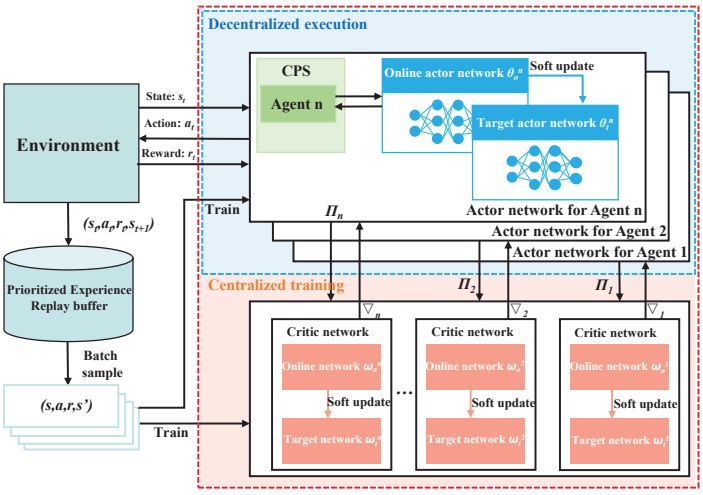


Fig. 2: The detailed diagram of PER-MADDPG.

The PER-MADDPG procedure is summarized in Algorithm 1. Target networks are updated by soft replacement with rate τ. This local learner provides the optimization backbone for each client in the federated framework.

D. FRL Training Algorithm

The federated component coordinates the subsystem agents without exchanging raw trajectories. Each agent is treated as a

client. To support heterogeneous local observations and actions, the policy of agent i is decomposed as

$$\pi_i(o_i) = g_{\xi_i}\big(b_\theta(e_{\psi_i}(o_i))\big), \tag{8}$$

where e_{ψ_i} is a local observation adapter, b_θ is the shared recovery backbone, and g_{ξ_i} is a local action and safety adapter. Only the backbone parameter θ is synchronized by the server. The adapter parameters ψ_i and ξ_i remain local because they encode subsystem-specific measurement channels, actuator dimensions, protection limits, and engineering constraints. When all agents share the same observation and action dimensions, these adapters reduce to identity or shallow linear maps, and the method becomes the standard shared-parameter AEFC-FRL used in the homogeneous case.

In each federation round, the server broadcasts the global backbone parameter vector θ^G and the latest clipped recovery prototypes defined below. Each client trains locally with PER-MADDPG, updates its local adapters and backbone copy, and uploads a sparse backbone update rather than raw trajectories or dense local policies. Fig. 3 shows this heterogeneous FRL architecture.

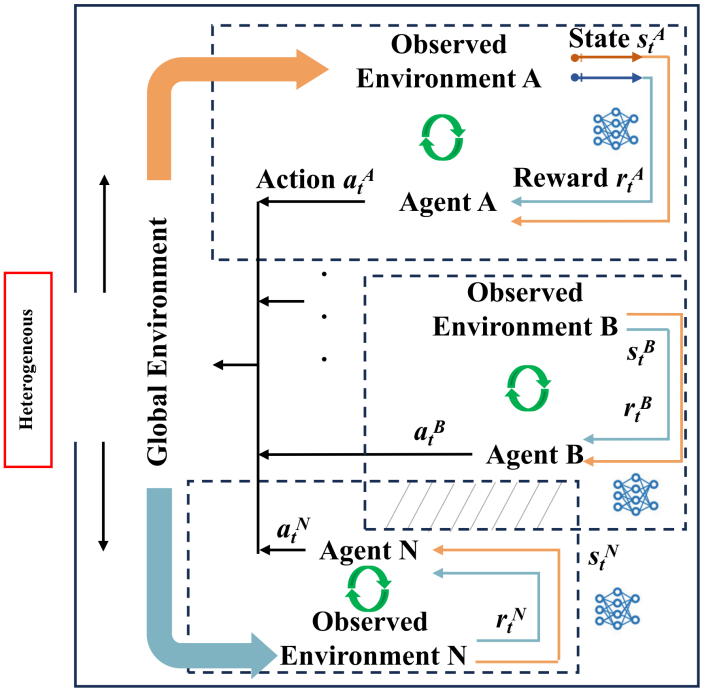


Fig. 3: The heterogeneous FRL architecture.

The setting is heterogeneous because agents may observe different local state variables and encounter different attack patterns. It is not vertical federated learning in the strict feature-partitioned sense. Instead, the framework assumes that local adapters map subsystem observations into a common latent recovery space. The shared backbone then learns recovery-control semantics that are common across agents, while input and output adapters absorb subsystem-specific dimensions. This decomposition avoids direct averaging of incompatible parameters and gives a concrete interface for extending the method to other CPS domains.

To further stabilize cross-client sharing, each client maintains a small set of recovery-state prototypes in the shared latent space. Let $z_i = e_{\psi_i}(o_i)$ and let $\mathcal{C} = \{\text{unsafe}, \text{recovering}, \text{stable}\}$ denote coarse recovery states determined from the safety violation $\phi(x_t, \mathcal{X}_{\text{safe}})$ and its one-step change. For category $m \in \mathcal{C}$, client i computes

$$p_{i,m} = \frac{1}{|\mathcal{B}_{i,m}|} \sum_{(o_i,s)\in\mathcal{B}_{i,m}} e_{\psi_i}(o_i), \tag{9}$$

where $\mathcal{B}_{i,m}$ is the recent local minibatch assigned to recovery state m. The server aggregates clipped prototypes by credibility-weighted averaging,

$$\bar{p}_m^t = \sum_{i=1}^{N} w_i^t \text{clip}(p_{i,m}, C_p). \tag{10}$$

These prototypes are not used as raw data surrogates; they provide a low-dimensional alignment signal for detecting domain or attack-context drift. During local training, the backbone is regularized by

$$L_{\text{align}}^i = \sum_{m\in\mathcal{C}} \frac{1}{|\mathcal{B}_{i,m}|} \sum_{(o_i,s)\in\mathcal{B}_{i,m}} \left\| e_{\psi_i}(o_i) - \bar{p}_m^{t-1} \right\|_2^2, \tag{11}$$

with the regularization weight chosen small enough that the safety reward remains the dominant learning signal. In local actor-backbone updates, Eq. 7 is therefore augmented as

$$L_{\pi_i}^{\text{total}} = L_{\pi_i} + \lambda_a L_{\text{align}}^i, \tag{12}$$

where $\lambda_a \geq 0$ controls the strength of recovery-state alignment. This term discourages representation drift across heterogeneous agents while preserving local personalization through the adapters.

For a basic federated round, client i computes an update $\Delta\theta_i = \theta_i^{(t)} - \theta_t^G$ after local training. A standard weighted aggregation can be written as

$$\theta_{t+1}^G = \theta_t^G + \eta_g \sum_{i=1}^{N} q_i \Delta\theta_i, \tag{13}$$

where $q_i \geq 0$, $\sum_i q_i = 1$, and η_g is the server learning rate. A common choice is $q_i = n_i / \sum_j n_j$, where n_i is the number of local training transitions used by agent i in the round. This baseline preserves data locality, but it can be communication-intensive and vulnerable to noisy or malicious updates. AEFC is designed to address these two weaknesses.

E. AEFC Optimization Mechanism

AEFC has three coupled components. The *Advantage Execution* component runs on each client and transmits only a sparse, high-utility backbone update. The *Recovery Alignment* component keeps local adapters personalized while aligning the shared latent recovery space through clipped prototypes. The *Federated Critic* component runs on the server and weights client updates according to a credibility score that combines recovery advantage, update scale, and shift consistency.

AEFC should be read as a recovery-aware composition rather than as a claim that sparsification, prototype alignment, or credibility weighting is new in isolation. Top-κ transmission is a standard way to reduce uplink cost, and trust weighting is common in robust aggregation. AEFC ties these operations to the semantics of resilient recovery: the client selects updates from high-advantage transitions that improve post-attack restoration, the shared backbone is aligned through recovery-state prototypes, and the server down-weights updates whose advantage signal is weak or whose representation drift is inconsistent with the global recovery state. This coupling is important in CPS because a large gradient is not necessarily a useful recovery signal under cross-layer attacks.

For client i, the advantage of a transition is estimated as

$$A_i(s,a) = Q_{\omega_i}(s,a) - V_i(s), \tag{14}$$

where $V_i(s)$ is the critic value under the current deterministic recovery actors,

$$V_i(s) = Q_{\omega_i}\big(s, \pi_{\theta_1}(o_1), \ldots, \pi_{\theta_N}(o_N)\big).$$

Transitions with larger positive advantage indicate recovery decisions that outperform the current local policy baseline and are therefore more useful for inter-client knowledge transfer.

Let

$$\mathcal{D}_i^{\text{high}} = \{(s, a, r, s') \in \mathcal{D}_i : A_i(s, a) > \zeta\} \tag{15}$$

be the high-advantage subset, where ζ is an advantage threshold. The local high-advantage loss is

$$\begin{aligned} L_i^{\text{high}}(\theta_i) = -\frac{1}{|\mathcal{D}_i^{\text{high}}|} \sum_{(s,a,r,s') \in \mathcal{D}_i^{\text{high}}} A_i(s, a) \\ \times Q_{\omega_i}\big(s, a_1, \ldots, \pi_{\theta_i}(o_i), \ldots, a_N\big), \end{aligned} \tag{16}$$

which is minimized as an advantage-weighted deterministic actor loss. The gradient of Eq. 16 uses the same deterministic policy-gradient form as Eq. 7, but it is computed only on the high-advantage subset. If a stochastic actor is used instead, the analogous surrogate can replace the critic term with $A_i(s, a) \log \pi_{\theta_i}(a_i|o_i)$. The implementation in this paper follows the deterministic MADDPG form.

After local training, client i forms a backbone update $\Delta\theta_i$ and applies Top-κ sparsification. Let $C_\kappa(\cdot)$ retain the largest $\lceil \kappa d \rceil$ entries in magnitude from a d-dimensional update and set all other entries to zero. The uploaded sparse update is

$$U_i = C_\kappa(\Delta\theta_i) = \{(j, \Delta\theta_{i,j}) : j \in \mathcal{I}_{i,\kappa}\}, \tag{17}$$

where $\mathcal{I}_{i,\kappa}$ is the selected index set. The client uploads only the index-value pairs, a scalar confidence statistic, and clipped recovery prototypes. Low-advantage or low-magnitude backbone changes remain local and are not communicated in that round. Adapter parameters are never uploaded.

The server computes a credibility score from the average selected advantage, a norm-control term, and a prototype-drift penalty. The drift of client i in round t is

$$d_i^t = \frac{1}{|\mathcal{C}|} \sum_{m \in \mathcal{C}} \left(1 - \frac{\langle p_{i,m}^t, \bar{p}_m^{t-1} \rangle}{\|p_{i,m}^t\|_2 \|\bar{p}_m^{t-1}\|_2 + \epsilon_p}\right), \tag{18}$$

where $\epsilon_p > 0$ avoids division by zero. Large d_i^t indicates that the client's latent recovery states have moved away from the previous global recovery semantics, which may reflect benign domain shift, stale training, or adversarial behavior. The credibility score is

$$\begin{aligned} c_i &= \frac{1}{|\mathcal{D}_i^{\text{high}}|} \sum_{(s,a) \in \mathcal{D}_i^{\text{high}}} A_i(s, a), \\ \rho_i &= \frac{\max(c_i, 0)}{\|U_i\|_2 + \epsilon_c} \exp(-\lambda_s d_i^t), \end{aligned} \tag{19}$$

where $\epsilon_c > 0$ avoids division by zero, $\lambda_s \geq 0$ controls shift sensitivity, and the exponential term suppresses updates whose latent recovery state is inconsistent with the global prototype memory. The aggregation weight is

$$w_i = \frac{\rho_i}{\sum_{j=1}^N \rho_j + \epsilon_c}. \tag{20}$$

The global update is then

$$\Delta\theta_t^G = \sum_{i=1}^N w_i U_i, \qquad \theta_{t+1}^G = \theta_t^G + \eta_g \Delta\theta_t^G. \tag{21}$$

In practice, the server also clips $\|U_i\|_2$ to a preset bound before aggregation. This prevents any single update from dominating the global model and improves robustness under poisoned or sign-flipped uploads.

Algorithm 2: FRL with Advantage Execution–Federated Critic (AEFC-FRL)

Input: Initial global backbone θ_0^G, number of agents N, total rounds R, local epochs E, sparsity ratio κ, prototype set $\mathcal{C}$

Output: Final global backbone θ^G and local adapters $\{(\psi_i, \xi_i)\}_{i=1}^N$

Initialize $\theta^G \leftarrow \theta_0^G$ and broadcast θ^G to all agents;
for $t = 0$ **to** $R - 1$ **do**
  for $i = 1$ **to** N **do**
    Set $\theta_i \leftarrow \theta_t^G$;
    Run E local epochs of PER-MADDPG (Alg. 1) and obtain $\theta_i^{(t)}$;
    Update local adapters (ψ_i, ξ_i) and compute recovery prototypes $\{p_{i,m}^t\}_{m \in \mathcal{C}}$;
    Estimate $A_i(s, a)$ on recent local transitions and form $\mathcal{D}_i^{\text{high}}$;
    Compute $\Delta\theta_i \leftarrow \theta_i^{(t)} - \theta_t^G$;
    Apply sparsification $U_i \leftarrow C_\kappa(\Delta\theta_i)$;
    Compute confidence statistic c_i and drift score d_i^t;
    Upload $(U_i, c_i, \{p_{i,m}^t\}_{m \in \mathcal{C}})$ to the server;
  Clip uploaded updates and compute credibility weights $\{w_i\}_{i=1}^N$ by Eq. 20;
  Update global recovery prototypes by Eq. 10;
  Aggregate $\Delta\theta_t^G \leftarrow \sum_{i=1}^N w_i U_i$;
  Update $\theta_{t+1}^G \leftarrow \theta_t^G + \eta_g \Delta\theta_t^G$ and broadcast θ_{t+1}^G;

AEFC reduces uplink traffic through sparse transmission and reduces the influence of unreliable clients through credibility-weighted aggregation. The prototype channel adds only $\mathcal{O}(|\mathcal{C}|r)$ real values per client per round, where r is the latent dimension, and is typically much smaller than the backbone update. Sparse gradients and prototypes, however, can still leak information about local trajectories, rare states, or high-impact recovery actions. Therefore, AEFC should be regarded as a communication-efficient and privacy-enhancing design rather than a formal privacy mechanism. A deployment that requires mathematical privacy guarantees can combine AEFC with secure aggregation or differential privacy. For example, client i may upload

$$\tilde{U}_i = C_\kappa(\Delta\theta_i) + \xi_i, \qquad \xi_i \sim \mathcal{N}(0, \sigma_{\text{dp}}^2 I), \tag{22}$$

where σ_{dp} is calibrated to a target privacy budget under a standard differential-privacy accountant. This extension trades privacy for recovery accuracy and is therefore evaluated separately from the main AEFC mechanism.

F. Theoretical Convergence Analysis

The convergence discussion below specifies the conditions under which a compressed distributed optimization argument is relevant to AEFC. It is not an unconditional convergence claim for arbitrary non-stationary MARL dynamics, nor does it imply global optimality of the learned recovery policy.

Let $F_i(\theta)$ be the expected local return objective of client i after fixing the replay distribution within a federation round, and let $F(\theta) = \sum_{i=1}^N q_i F_i(\theta)$ with $q_i \geq 0$ and $\sum_i q_i = 1$. Because the RL objective is non-convex, convergence is measured by the expected squared gradient norm. We use the following assumptions.

Assumption 1 (Smoothness)**.** *Each F_i is L-smooth, i.e., $\|\nabla F_i(\theta) - \nabla F_i(\theta')\| \le L\|\theta - \theta'\|$ for all θ, θ'.*

Assumption 2 (Stochastic Gradient Noise)**.** *The local stochastic gradient g_i^t satisfies $\mathbb{E}[g_i^t|\theta_t] = \nabla F_i(\theta_t)$ and $\mathbb{E}\|g_i^t - \nabla F_i(\theta_t)\|^2 \le \sigma^2$.*

Assumption 3 (Bounded Heterogeneity)**.** *There exists $G_h \ge 0$ such that $\sum_{i=1}^N q_i \|\nabla F_i(\theta) - \nabla F(\theta)\|^2 \le G_h^2$ for all θ visited during training.*

Assumption 4 (Contractive Compression)**.** *The sparsification operator satisfies $\mathbb{E}\|C_\kappa(v) - v\|^2 \le (1 - \delta_\kappa)\|v\|^2$ for some $\delta_\kappa \in (0, 1]$. For Top-κ compression, δ_κ depends on the retained fraction and the concentration of the update vector.*

Assumption 5 (Bounded Credibility Weights)**.** *The server weights satisfy $0 \le w_i^t \le w_{\max}$ and $\|\sum_i (w_i^t - q_i)\nabla F_i(\theta_t)\| \le b_w$. The term b_w captures the bias introduced by advantage-based and shift-aware weighting. The prototype-drift scores are bounded by $0 \le d_i^t \le D_s$ after clipping.*

Theorem 1 (Conditional stationarity bound)**.** *Suppose Assumptions 1–5 hold for the round-level surrogate objective defined above, and let the AEFC update be written as*

$$\theta_{t+1} = \theta_t + \eta_t \sum_{i=1}^N w_i^t C_\kappa(g_i^t). \tag{23}$$

If $F(\theta) \le F^\star$ for all iterates and the local gradients are bounded by G, then, for a constant step size η small enough for the smoothness term to be controlled, the iterates satisfy

$$\frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}\|\nabla F(\theta_t)\|^2 \le \mathcal{O}\left(\frac{F^\star - F(\theta_0)}{\eta T} + \eta(\sigma^2 + G_h^2) + (1 - \delta_\kappa)G^2 + b_w^2\right), \tag{24}$$

where $(1 - \delta_\kappa)G^2$ is the residual error caused by sparse compression and b_w^2 is the residual bias caused by credibility-aware weighting.

Proof. Since each F_i is L-smooth, F is also L-smooth. Applying the smoothness inequality for gradient ascent gives $F(\theta_{t+1}) \ge F(\theta_t) + \eta_t\langle \nabla F(\theta_t), \tilde{g}_t\rangle - \frac{L\eta_t^2}{2}\|\tilde{g}_t\|^2$, where $\tilde{g}_t = \sum_i w_i^t C_\kappa(g_i^t)$ is the aggregated sparse update. Add and subtract $\sum_i q_i \nabla F_i(\theta_t)$ inside the inner product. The stochastic-gradient term is bounded by Assumption 2, the inter-client drift by Assumption 3, the sparsification residual by Assumption 4, and the credibility-weighting bias by Assumption 5. Summing the resulting inequality from $t = 0$ to $T - 1$, using $F(\theta_T) \le F^\star$, and dividing by T yields Eq. 24. This proof follows the usual non-convex compressed federated optimization template, but it keeps the AEFC-specific compression and credibility-bias terms explicit instead of absorbing them into the SGD variance. □

Eq. 24 should be interpreted as a conditional stationarity bound. If $\eta = \Theta(T^{-1/2})$ and the residual compression and weighting terms are controlled, the first two terms scale as $\mathcal{O}(T^{-1/2})$. The complete bound, however, also contains $(1 - \delta_\kappa)G^2$ and b_w^2. Therefore AEFC is not claimed to converge at a pure $\mathcal{O}(T^{-1/2})$ rate unless these residuals are negligible or decrease with training.

Two qualifications are necessary. First, MARL creates non-stationary data distributions because each agent's policy changes during learning; the argument applies to the round-level surrogate objective where the replay distribution is treated as locally fixed. Second, Top-κ compression, advantage weighting, and shift-aware prototype penalties are biased operations. Their effect is explicitly represented by $(1 - \delta_\kappa)G^2$ and b_w^2 rather than assumed away. Section IV reports training diagnostics that monitor whether these residual terms remain controlled in the studied power-grid CPS benchmarks.

Training Diagnostics. The following quantities are recorded during training to connect the assumptions with empirical behavior:

$$\begin{aligned} G_t &= \left\| \sum_{i=1}^N w_i^t g_i^t \right\|_2, \\ H_t &= \sum_{i=1}^N q_i \|g_i^t - \bar{g}^t\|_2^2, \\ B_t &= \left\| \sum_{i=1}^N (w_i^t - q_i) g_i^t \right\|_2, \\ S_t &= \frac{1}{N}\sum_{i=1}^N d_i^t, \end{aligned} \tag{25}$$

where $\bar{g}^t = \sum_i q_i g_i^t$. Here, G_t tracks the aggregated gradient norm, H_t estimates inter-client gradient heterogeneity, B_t measures the bias introduced by credibility weighting, and S_t measures latent recovery-state drift. Stable or decreasing G_t, bounded H_t, small B_t, and non-explosive S_t provide empirical support for the assumptions used in Eq. 24. These diagnostics do not prove convergence by themselves, but they make the assumptions auditable rather than purely declarative.

Communication Complexity. In standard federated RL, each of the N clients uploads a dense d-dimensional update in every round, giving uplink complexity $\mathcal{O}(NdT)$ over T rounds. AEFC uploads only κd index-value pairs per client and $|\mathcal{C}|r$ prototype values, so the uplink complexity becomes $\mathcal{O}(N(\kappa d + |\mathcal{C}|r)T)$, plus a small scalar confidence statistic per client. Because $|\mathcal{C}|r \ll d$ in the studied models, the dominant term remains $\mathcal{O}(N\kappa dT)$. The downlink broadcast remains $\mathcal{O}(NdT)$ if each client receives a dense global backbone, although practical server-to-client broadcast is often less restrictive than the many-to-one uplink. Thus, AEFC primarily reduces the uplink bottleneck, and the reduction is controlled by the sparsity ratio κ with a small prototype-alignment overhead.

IV. EXPERIMENTAL

The evaluation focuses on power-grid CPS recovery simulations and examines recovery performance, communication cost, heterogeneity stress, robustness against malicious clients, membership-inference risk, scalability, convergence diagnostics, and deployment issues. A small non-grid traffic-signal proof of concept is also included to test whether the AEFC interface depends on power-flow equations. The source code is available at https://github.com/startkz/AEFC-FRL.

A. Experiment Setup

All experiments were conducted on a workstation with an Intel(R) Xeon(R) Gold 6230 CPU at 2.10 GHz, 128 GB RAM, and an NVIDIA RTX 3090 GPU. The software environment used Ubuntu 20.04 LTS, Python 3.9, PyTorch 2.0, and CUDA acceleration. The CPS environment was built on a Simulink–Python co-simulation platform: Simulink models the physical power-grid dynamics, and a custom Gym-compatible interface exposes states, actions, and rewards to the learning agents.

The framework is evaluated on two *power-grid CPS* benchmarks: the IEEE 39-bus system and the IEEE 118-bus system. The 39-bus case is used for convergence, recovery performance, robustness, and membership-inference evaluation. The 118-bus case is used to test scalability with different numbers of agents. The current empirical evidence is therefore strongest for power-grid recovery. For other CPS domains, such as transportation control or industrial automation, the state-action semantics, actuation delay, safety constraints, and topology can differ substantially; those differences are treated as external-validity limitations rather than as already validated generalization.

The IEEE 39-bus system contains 10 generators and 39 buses and is partitioned into multiple subsystems for multi-agent control. Fig. 4 shows the co-simulation platform, where the physical network, subsystem agents, and federated coordination server are connected in one recovery pipeline. Each agent observes local measurements from its assigned subsystem and executes recovery actions under attack conditions. In federated variants, the server aggregates model updates rather than raw trajectories.

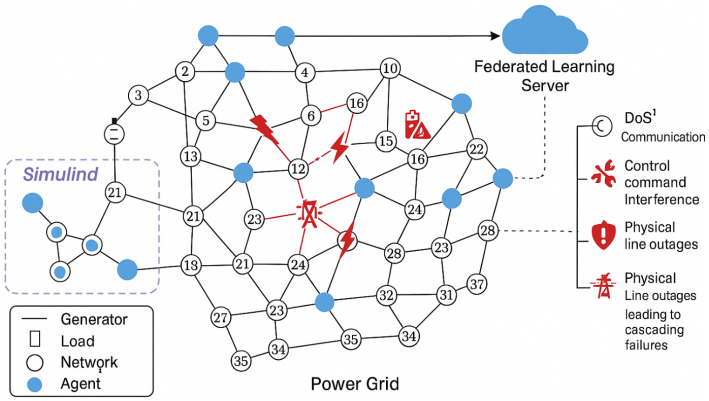


Fig. 4: Simulink–Python co-simulation platform and federated recovery pipeline.

All learning methods use the same MADDPG-based actor–critic backbone. Unless otherwise stated, the actor and critic learning rates are both 0.001 with Adam, the replay buffer size is 1×10^5 transitions per agent, the batch size is 256, the discount factor is 0.99, and the soft target update rate is $\tau = 0.005$. Each actor and critic has two hidden layers with 128 units per layer and ReLU activations. Exploration uses Ornstein–Uhlenbeck noise with an initial scale of 0.2 that decays during training. Federated synchronization is performed every 5 episodes. For AEFC, the sparsification ratio is $\kappa = 0.2$. All methods are trained for 1000 episodes with 30 steps per episode. Table I summarizes the shared hyperparameter settings.

TABLE I: Parameter settings

Parameter	Value
Learning rate (actor/critic)	0.001
Discount factor γ	0.99
Replay buffer size	1×10^5 transitions/agent
Batch size	256
Target update rate τ	0.005
Exploration noise coefficient	0.15
Federation synchronization interval	5 episodes
Gradient sparsification ratio κ	0.2
Hidden layers (actor/critic)	2 layers $\times$ 128 units
Training episodes $\times$ steps/episode	1000 $\times$ 30

We compare AEFC-FRL with the following baselines and ablation variants.

1) *MADDPG*: a centralized learner that trains a single controller using information from the whole 39-bus system. This is an idealized full-information baseline.

2) *FedRL*: a standard federated multi-agent RL baseline. Each agent trains locally and periodically uploads dense model updates for server averaging. It preserves data locality but does not use advantage-guided sparsification or credibility-aware aggregation.

3) *FedRL-NoAdv*: an ablation variant without advantage execution. Clients use credibility-aware aggregation but upload dense updates, which isolates the contribution of sparse advantage-guided transmission.

4) *FedRL-NoCW*: an ablation variant without critic weighting. Clients use advantage-guided sparse updates, while the server applies equal-weight aggregation.

B. Performance Evaluation

Fig. 5 reports recovery time over training episodes. At the beginning of training, all methods require roughly 15–18 time steps to stabilize the system after an attack. As training proceeds, recovery time decreases for all methods. AEFC-FRL reaches a stable level of about 9 recovery steps after approximately 300 episodes. MADDPG and FedRL converge more slowly and settle around 13 and 12 steps, respectively. FedRL-NoAdv and FedRL-NoCW fall between the full method and the baselines: FedRL-NoAdv plateaus at about 10–11 steps, while FedRL-NoCW eventually reaches about 9 steps but needs more training episodes. These results indicate that sparse advantage execution and credibility-aware aggregation improve both recovery speed and training stability in the studied 39-bus setting.

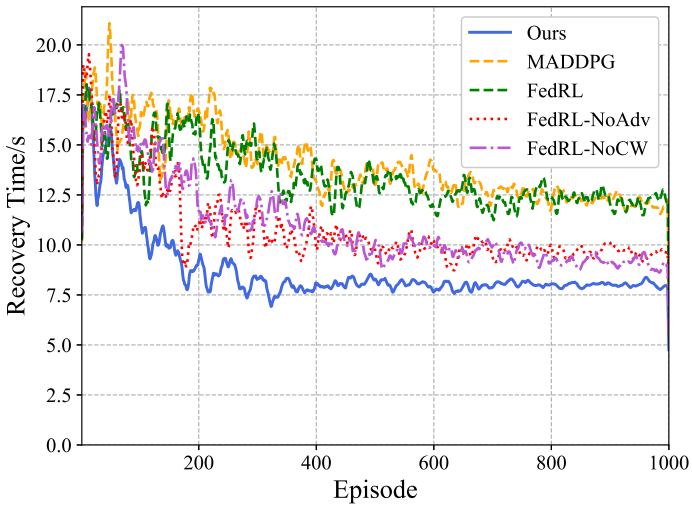


Fig. 5: Convergence of recovery time.

Fig. 6 shows recovery accuracy, measured by the percentage of restored load or the fraction of state variables returned to safe ranges after an attack. All methods start at about 60%–70% because the initial policies are nearly random. AEFC-FRL rises above 90% within the first 200 episodes and converges near 98%. MADDPG plateaus around 85%, and FedRL reaches about 88%. FedRL-NoAdv and FedRL-NoCW converge to roughly 93% and 94%, respectively. The final gain of AEFC-FRL over centralized MADDPG is about 15.3 percentage points. The ablation results show that both components of AEFC contribute: advantage execution improves the quality and efficiency of transmitted updates, while critic weighting improves global aggregation.

Fig. 7 compares communication overhead during training for the distributed methods. AEFC-FRL transmits about 270 KB per

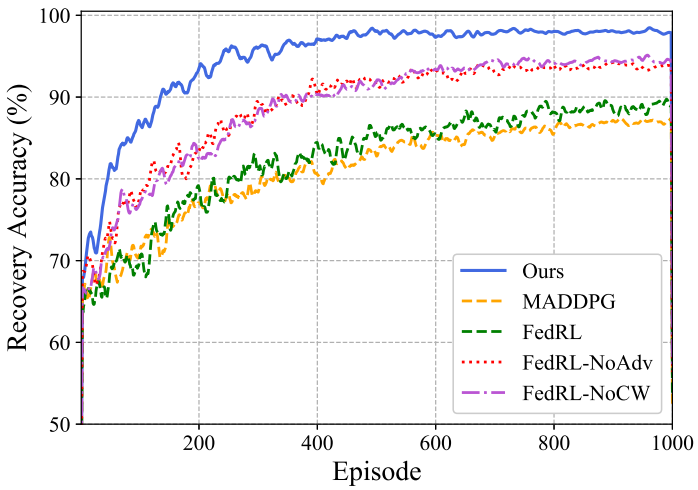


Fig. 6: Convergence of recovery accuracy.

episode, whereas FedRL requires about 500 KB per episode. This corresponds to a 45.8% reduction in communication overhead. FedRL-NoCW has a similar communication profile to AEFC-FRL because both use sparse uploads. FedRL and FedRL-NoAdv upload dense updates and remain around 480–510 KB per episode. These results confirm that the communication saving mainly comes from advantage-guided sparsification, while the accuracy gain depends on combining sparsification with credibility-aware aggregation.

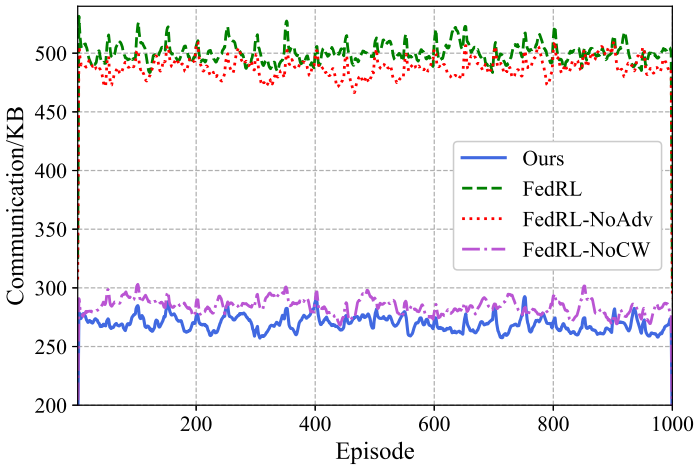


Fig. 7: Communication overhead during federated training.

C. Hyperparameter Sensitivity and Runtime Cost

AEFC introduces two practical hyperparameters: the advantage threshold ζ and the sparsification ratio κ. The threshold ζ controls which local transitions are treated as high-value recovery experiences, while κ controls the number of update coordinates transmitted to the server. A very low ζ or large κ makes AEFC close to dense FedRL and weakens the communication benefit. A very high ζ or small κ can discard useful recovery signals and slow policy improvement. For this reason, the default setting uses a moderate sparsification ratio $\kappa = 0.2$ and selects high-advantage transitions from the upper range of the local advantage distribution.

In deployment, these parameters need not remain fixed. The advantage threshold can be set adaptively as a local quantile, e.g., selecting the top $p\%$ of recent advantage values in each client buffer. Similarly, κ can be adjusted according to available uplink bandwidth or server-side congestion. When the channel is congested, clients can reduce κ and upload fewer index-value pairs; when recovery performance drops or attacks intensify, κ can be relaxed to transmit more information. This adaptive interpretation makes AEFC less dependent on hand-tuned constants and provides a practical path for bandwidth-aware operation.

AEFC also adds computation beyond standard FedRL. The main additional costs are advantage scoring, Top-κ coordinate selection, norm clipping, and credibility-weight calculation. Advantage scores are already available from the actor–critic learner, so the dominant extra client-side operation is Top-κ selection over the update vector. This operation is linear or near-linear in the number of parameters with standard partial-selection routines and is executed only at federation rounds rather than at every environment step. The overhead is therefore mainly a synchronization-round cost, while the saved communication volume is paid back whenever uplink bandwidth is the bottleneck. For real-time CPS, the federation interval should be chosen so that all AEFC processing finishes outside the hard control loop, with certified local controllers handling immediate safety responses.

D. Heterogeneity and Shift Stress Test

We further evaluate whether the adapter and prototype-alignment components improve stability when subsystem agents no longer share perfectly aligned observations and attack contexts. Four heterogeneity levels are considered. H0 uses aligned local observations and identical adapter dimensions. H1 masks 20% of local sensing channels for a subset of clients. H2 combines sensing masks with communication delay and separates clients into sensor-attack and control-attack groups. H3 increases the sensing mask to 40%, adds stochastic message delay, and applies actuator saturation to part of the system. The same federation interval, sparsification ratio, and local training budget are used across all levels.

Fig. 8 reports recovery accuracy averaged over five seeds. Standard FedRL degrades sharply as heterogeneity increases, falling from about 88% at H0 to about 64% at H3. AEFC without local adapters performs better than FedRL but still loses accuracy when observation and action mappings become incompatible. Removing recovery-state prototypes also hurts performance under H2 and H3, showing that sparse advantage updates alone do not fully control latent drift. The full AEFC-FRL maintains about 88% recovery accuracy at H3, which is approximately 9 percentage points higher than the no-adapter variant and 24 percentage points higher than FedRL. This stress test supports the design choice of synchronizing an aligned recovery backbone while keeping subsystem-specific adapters local.

E. Robustness and Empirical Privacy-Risk Assessment

Robustness is evaluated by injecting Byzantine clients into the federated training process at ratios of 10%, 20%, and 30%. The malicious clients submit random updates or poisoned gradients, including scaled and sign-flipped perturbations. Fig. 9 reports the resulting recovery accuracy. As the fraction of malicious clients increases, all methods degrade, but AEFC-FRL degrades more slowly than the unenhanced federated variants. At 30% malicious participation, AEFC-FRL preserves approximately 83% recovery accuracy. This behavior is consistent with the design goal of AEFC: advantage-guided filtering reduces the spread of low-utility local updates, and credibility-aware aggregation lowers the weight assigned to suspicious contributions.

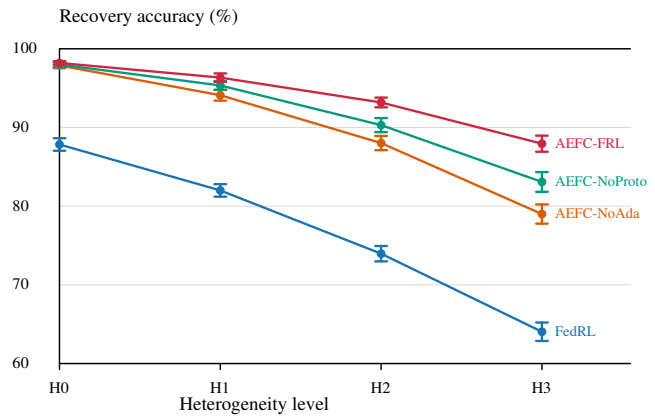


Fig. 8: Recovery accuracy under increasing subsystem heterogeneity.

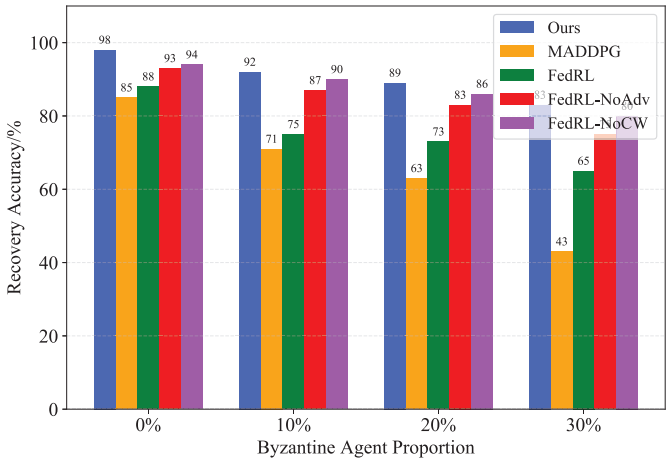


Fig. 9: Recovery accuracy under different Byzantine-client ratios.

Privacy risk is evaluated through a membership inference attack (MIA). The adversary attempts to determine whether a target sample belongs to a client's local training set. Attack success is measured by AUC, where an AUC close to 0.5 indicates performance close to random guessing under the adopted attack model. Fig. 10 shows that AEFC-FRL produces the lowest AUC among the tested methods and approaches 0.5.

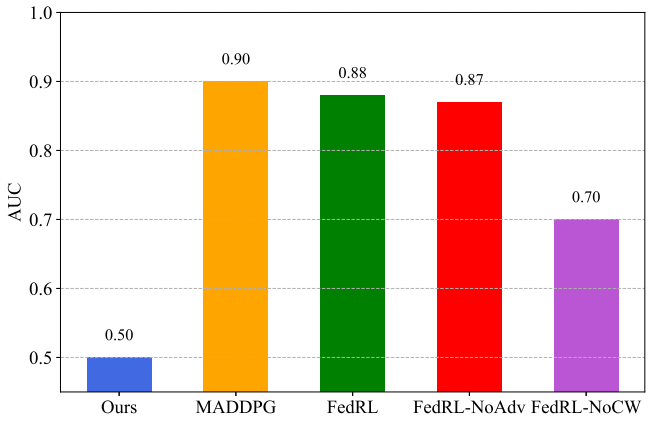


Fig. 10: Membership inference attack AUC for each method.

A near-0.5 AUC in this experiment indicates reduced empirical membership-inference efficacy under the chosen attacker and data split. It is not a formal privacy guarantee, and it does not imply resistance to stronger inference, reconstruction, or gradient-leakage attacks. AEFC reduces the amount of transmitted information, but sparsification alone is not a substitute for differential privacy, secure aggregation, or cryptographic protection. The reported result is therefore an empirical privacy-risk assessment without formal guarantees, rather than proof of privacy preservation.

Gradient leakage is a separate threat. A gradient-reconstruction adversary may use uploaded index-value pairs, model snapshots, and prior knowledge of the subsystem dynamics to infer sensitive local states or rare attack trajectories. AEFC makes such reconstruction harder by transmitting fewer coordinates, but the nonzero coordinates are also more informative because they are selected from high-advantage transitions. In a high-assurance deployment, sparse uploads should therefore be combined with secure aggregation, update clipping, and calibrated differential-privacy noise. The expected trade-off is clear: stronger formal privacy mechanisms reduce reconstruction risk but may also attenuate the recovery signal, especially when high-advantage events are rare.

Fig. 11 complements the Byzantine analysis by examining performance degradation under three attack-intensity levels relative to the no-attack case. AEFC-FRL remains stable across the three regimes. The degradation is small under low intensity, moderate under medium intensity, and remains bounded under high intensity. The result supports the claim that sparse update transmission and credibility-aware aggregation should be used together: sparsification reduces communication and filters weak updates, while credibility weighting improves resistance to outliers.

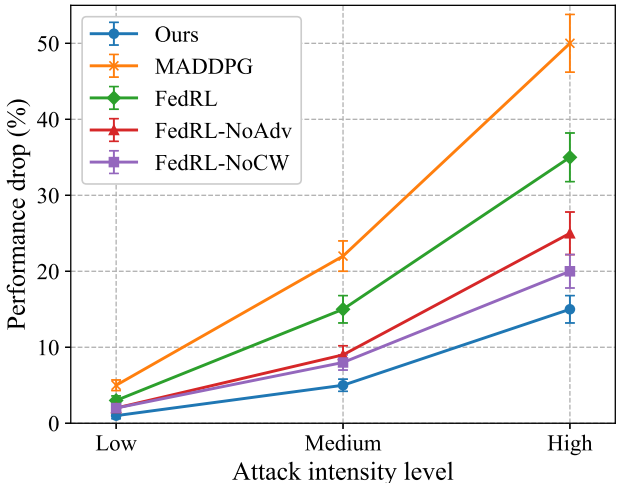


Fig. 11: Robustness comparison under different attack intensities.

F. Scalability Evaluation

Scalability is evaluated on the IEEE 118-bus system with 10, 20, 30, and 40 agents. As shown in Fig. 12, AEFC-FRL reaches 90% recovery accuracy in about 200–250 episodes across the tested agent counts. MADDPG and FedRL plateau below the 90% threshold in the same training budget, while FedRL-NoAdv and FedRL-NoCW eventually exceed 90% but require more episodes. The advantage of AEFC-FRL becomes more visible as the number of agents increases, suggesting that federated knowledge sharing is useful when the grid is partitioned into many local controllers.

Fig. 13 reports final recovery accuracy under different numbers of agents. AEFC-FRL remains near 95%–98% restored load across the tested scales. MADDPG is lower and becomes less stable as the number of agents increases, dropping to roughly 70%–80% at 40 agents. FedRL remains below 90% in most cases.

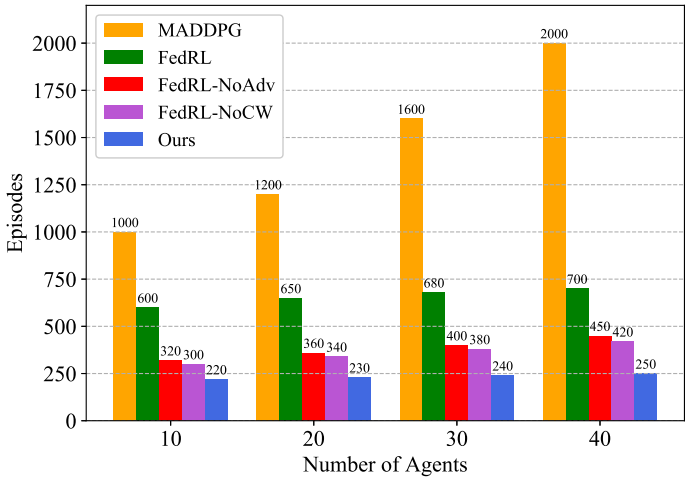


Fig. 12: Episodes required to reach the target recovery accuracy under different numbers of agents.

FedRL-NoAdv and FedRL-NoCW reach about 93%–95%, but remain below the full AEFC-FRL method. These results indicate that the two AEFC components remain useful when the system size and number of local controllers increase.

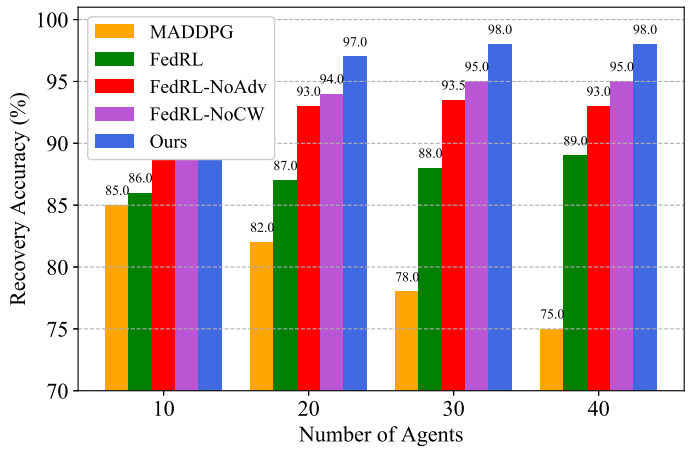


Fig. 13: Final recovery accuracy under different numbers of agents.

Fig. 14 shows communication overhead on the 118-bus system. With 10 agents, AEFC-FRL requires about 270 KB per round on average, compared with about 500 KB for FedRL. As the number of agents increases, total communication grows approximately linearly. At 40 agents, FedRL reaches about 2,000 KB per round, while AEFC-FRL uses about 1,080 KB. Across the tested scales, AEFC-FRL reduces communication by approximately 45%–46% relative to standard FedRL.

G. Convergence Diagnostics

The convergence analysis in Section III relies on bounded update norm, bounded inter-client heterogeneity, controlled bias from credibility weighting, and bounded prototype drift. To make these assumptions auditable in implementation, four diagnostic quantities are logged during training: the aggregated update norm G_t, the relative heterogeneity estimate H_t, the relative credibility-weighting bias B_t/G_t, and the prototype-drift score S_t. Their theoretical definitions are given in Section III-E. Because the released actor–critic implementation exposes client parameter deltas after local optimizer steps rather than raw stochastic

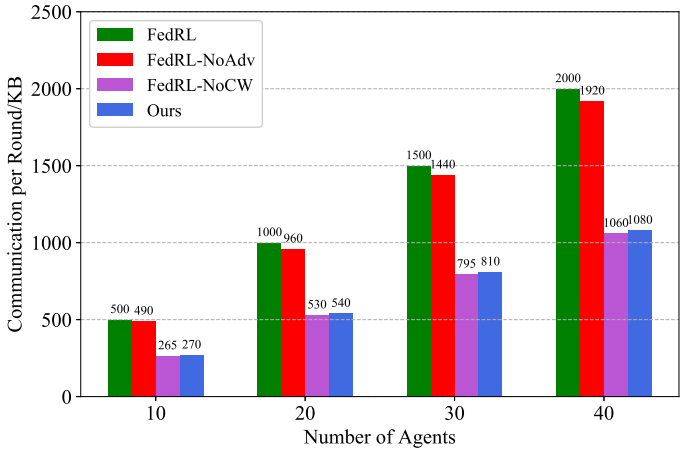


Fig. 14: Communication overhead under different numbers of agents.

gradients, the logged values in Table II use the corresponding round-level update proxies. We instrumented the released AEFC-FRL reference implementation on a CPS-like Pendulum-v1 stabilization task with five clients, 10% random Byzantine clients, Top-$\kappa = 0.2$, and three random seeds. This lightweight run is used only to audit the training assumptions; it is not reported as an additional power-grid benchmark.

TABLE II: Measured diagnostics for auditing convergence assumptions

Metric	Audited quantity	Mean ± std.	P95	Max
G_t	$\|\Delta_t\|_2$	2.092 ± 0.309	2.580	2.663
H_t	Relative client heterogeneity	1.129 ± 0.138	1.313	1.324
B_t/G_t	Relative weighting bias	0.032 ± 0.015	0.060	0.066
S_t	Prototype-drift score	0.022 ± 0.011	0.039	0.055

Values are computed over 21 post-warm-up federation rounds from the released reference implementation. Update-based diagnostics are used as executable proxies for the gradient-based quantities in Section III-E. They quantify whether the boundedness assumptions remain empirically plausible; they are not a formal convergence proof.

These diagnostics are used as sanity checks rather than as substitutes for proof. When G_t decreases together with recovery loss, the optimization process is consistent with the stationarity criterion in Eq. 24. When H_t, B_t/G_t, or S_t spikes, the corresponding federation round is inspected for local attack imbalance, representation drift, or abnormal client updates. This practice is useful in deployment because it exposes when the assumptions behind the convergence discussion cease to be reliable.

H. External Validity and Deployment

The power-grid benchmarks exercise important CPS properties, including continuous dynamics, coupled subsystems, safety constraints, and communication-limited distributed control. Nevertheless, they do not cover every CPS domain. To test whether AEFC is tied to the specific structure of power-flow equations, we add a small non-grid proof of concept using a two-phase traffic-signal queue simulator. Each client represents one local intersection with different arrival rates. A cross-layer incident injects sensor bias and command delay during the recovery window. The local learner uses a lightweight policy-gradient actor with a local observation encoder, and the server receives either dense FedRL updates or AEFC sparse updates selected by recovery advantage. The same interface is preserved: local observation encoding, local policy update, sparse upload, and server aggregation. Table III reports this proof-of-concept result.

TABLE III: Non-grid traffic-signal proof of concept under sensor-bias and command-delay attacks

Method	Avg. queue ↓	Median queue [IQR] ↓	Recovery acc. (%) ↑	Comm. units ↓
FedRL	35.19	44.06 [13.05, 46.92]	45.55	4800
FedRL-Clip	24.43	22.53 [13.60, 31.90]	55.47	4800
AEFC-FRL	29.03	33.89 [11.74, 45.29]	47.25	3200

The table uses 15 random seeds. The seed-level distribution is skewed, so the table reports both the mean queue length and the median queue with interquartile range; full mean–standard-deviation values are retained in the released CSV logs. AEFC-FRL reduced mean queue length relative to dense FedRL while using one-third fewer communication units. A dense clipped baseline performed best on this small traffic simulator, which shows that the non-grid result should be interpreted as proof of interface portability rather than as a claim of traffic-control superiority. The result supports the narrower claim that AEFC's local-encoder, sparse-upload, and aggregation interface is not hard-coded to grid topology.

The comparison with CyberForce in Table IV also clarifies the robustness boundary. Under 30% malicious clients, CyberForce reports 95.0%–98.0% recovery accuracy, whereas AEFC-FRL preserves 83.0%. AEFC-FRL therefore should not be read as replacing specialized Byzantine-defense mechanisms under severe adversarial participation. Its design trades some worst-case robustness for lower communication cost and recovery-conditioned sharing. In deployments where the expected malicious-client ratio is high, AEFC should be combined with stronger robust aggregation, anomaly screening, or secure execution mechanisms.

The privacy boundary is similarly important. AEFC does not include differential privacy, secure aggregation, or a cryptographic protocol. Sparse high-advantage updates can reduce the number of exposed coordinates, but the coordinates that remain may be especially informative because they are selected from rare recovery-critical transitions. Such updates may become high-value targets for gradient-reconstruction attacks, especially when an adversary has access to public model snapshots and prior knowledge of local dynamics. A high-assurance deployment should combine AEFC with secure aggregation, update clipping, calibrated differential-privacy noise, or TEE-based protection. These additions may reduce recovery signal strength, so the privacy–resilience trade-off needs to be evaluated explicitly.

The main empirical conclusion remains bounded: AEFC-FRL has been extensively validated on power-grid CPS and lightly checked on a non-grid traffic-signal simulator. Broader CPS validation still requires domain-specific simulators, safety metrics, actuation-delay models, and fallback controllers.

Practical deployment also raises engineering constraints that are not fully captured by simulation. Federated rounds require clock synchronization or an asynchronous aggregation policy when edge devices report at different rates. Edge controllers may not have enough memory or compute capacity for large actor–critic models, making model compression and local update scheduling necessary. Communication links can be intermittent, so stale updates should be timestamped, clipped, or discarded. The additional runtime cost of advantage scoring and Top-κ selection should be scheduled outside the hard real-time control period. Finally, operational CPS often require fail-safe fallback controllers; AEFC should therefore be deployed as an adaptive recovery layer that remains subordinate to certified protection logic during unsafe transients.

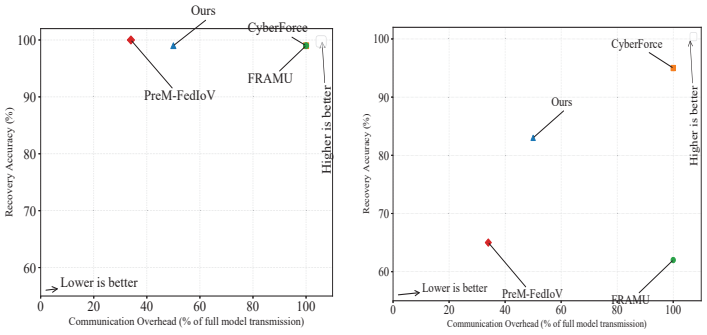
I. Comparative Evaluation

AEFC-FRL is also compared with three representative federated learning or federated reinforcement learning methods: CyberForce [49], FRAMU [50], and PreM-FedIoV [51]. Table IV summarizes the comparison under 10% and 30% malicious participation.

TABLE IV: Comparison with representative federated methods under 10% and 30% Byzantine clients

Method	**Convergence Rounds**		**Recovery Accuracy (%)**		**Comm. Cost (% of FedRL)**	
	10%	**30%**	**10%**	**30%**	**10%**	**30%**
AEFC-FRL (Ours)	∼200	∼220	**99.0**	83.0	**50.0**	**50.0**
CyberForce	∼200	∼210	98.0–100.0	**95.0–98.0**	100.0	100.0
FRAMU	<200	>250	98.0–100.0	∼60.0–65.0	100.0	100.0
PreM-FedIoV	∼230	>260	100.0	∼65.0	34.0	34.0

The comparison reveals a trade-off. CyberForce is the most robust under heavy malicious participation, but it incurs full communication cost. PreM-FedIoV is more communication-efficient, with communication cost reported as 34.0% of standard FedRL under both malicious-client ratios, yet its recovery performance deteriorates under adversarial participation. AEFC-FRL occupies a middle point for the target recovery setting: it preserves high recovery accuracy under moderate malicious participation while cutting communication roughly in half. The method does not dominate every baseline on every metric, but it provides a favorable efficiency–robustness balance for the studied power-grid CPS recovery tasks.



(a) 10% malicious clients (b) 30% malicious clients

Fig. 15: Pareto trade-off curves of recovery accuracy versus communication overhead under different Byzantine-client ratios.

The Pareto view in Fig. 15 shows the same pattern. Under moderate adversarial participation, communication-efficient methods remain competitive. Under heavier Byzantine participation, robustness becomes more important, and AEFC-FRL and CyberForce remain near the Pareto frontier. This comparison supports the main design choice of AEFC-FRL: sparse communication and robustness-aware aggregation should be optimized jointly rather than treated as separate add-ons.

V. CONCLUSION

This paper presented a security-aware FRL framework for collaborative resilient recovery in distributed CPS under cross-layer attacks. The framework combines subsystem-level multi-agent reinforcement learning with an Advantage Execution–Federated Critic mechanism, enabling local recovery policies to share useful knowledge without exchanging raw trajectories. The design targets three requirements that are central to attack-time CPS recovery: sharing only an aligned recovery backbone across heterogeneous subsystems, transmitting recovery-informative updates over constrained links, and reducing the influence of unreliable or malicious client updates.

The empirical evaluation on the IEEE 39-bus and IEEE 118-bus power-grid benchmarks showed that the proposed method can reduce recovery time and communication overhead while maintaining resilience under subsystem heterogeneity and adversarial participation. The heterogeneity stress test further showed that local adapters and recovery-state prototypes help preserve recovery accuracy when observations, actuation constraints, and attack contexts differ across agents. The privacy-risk evaluation based on membership inference attacks suggests reduced attack efficacy under the tested protocol, but it is not a formal privacy guarantee. Sparse updates and clipped prototypes may still leak sensitive information through gradient reconstruction, and deployment in privacy-critical CPS should incorporate secure aggregation, clipping, and differential privacy when required.

The main validation remains concentrated on power-grid CPS, with only a small traffic-signal proof of concept used to check interface portability. Extension to full traffic-control benchmarks, industrial PLC networks, and other CPS domains requires domain-specific state encoders, safety metrics, actuation-delay models, and fallback controllers. Practical deployment also requires careful treatment of federated-round synchronization, edge-device compute limits, intermittent communication, stale updates, robust aggregation under high malicious-client ratios, and certified safety logic. These issues define the next step from simulation-grade resilient recovery toward deployable secure learning in real CPS.

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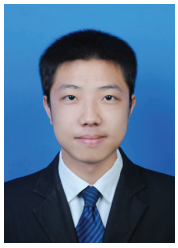
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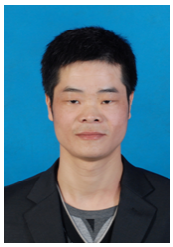
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